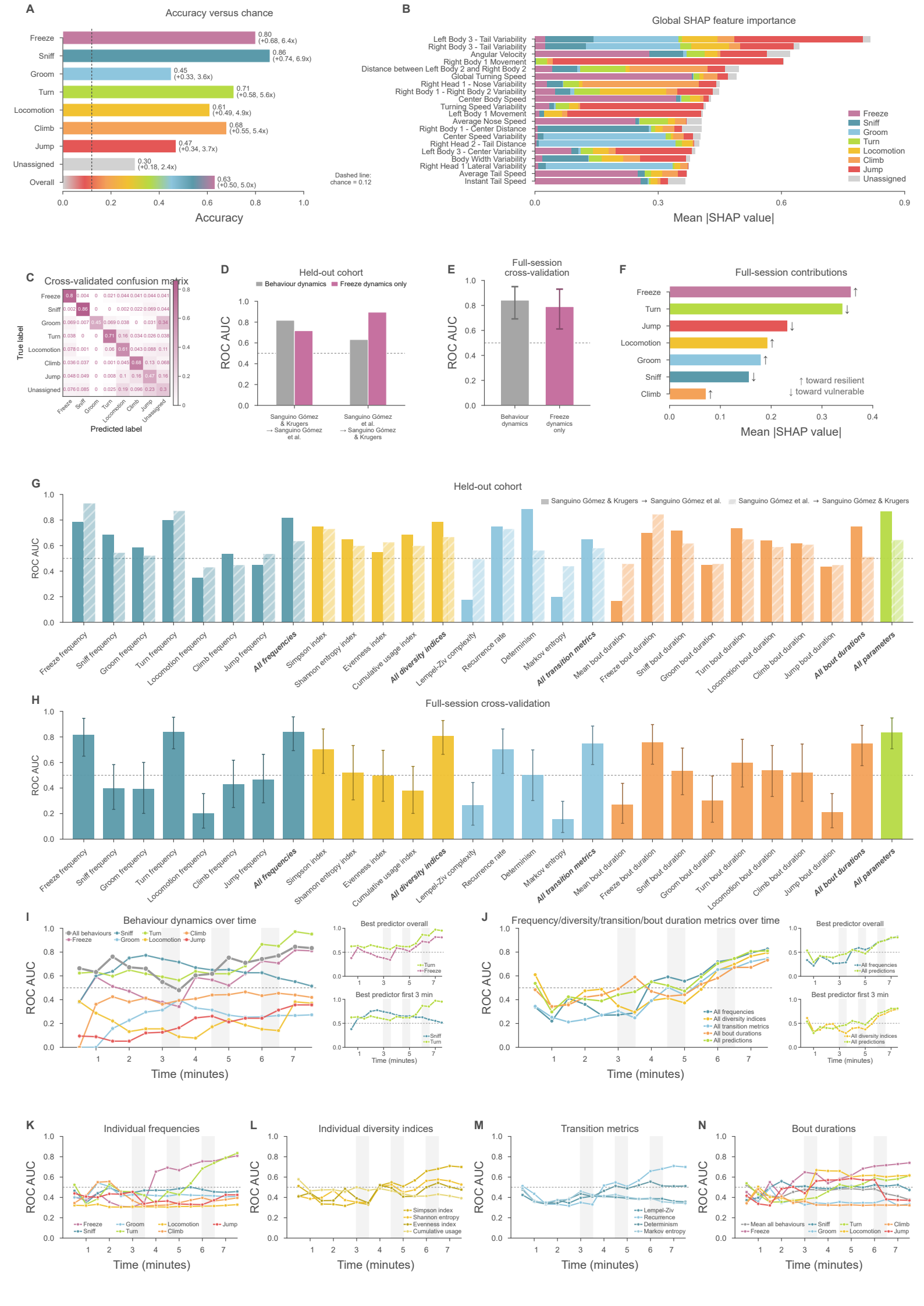

A
Accuracy versus chance
Freeze
Sniff
Groom
Turn
Locomotion
Climb
Jump
Unassigned
Overall
0.80 (+0.68, 6.4x)
0.86 (+0.74, 6.9x)
0.45 (+0.33, 3.6x)
0.71 (+0.58, 5.6x)
0.61 (+0.49, 4.9x)
0.68 (+0.55, 5.4x)
0.47 (+0.34, 3.7x)
0.30 (+0.18, 2.4x)
0.63 (+0.50, 5.0x)
Dashed line: chance = 0.12
Accuracy
B
Global SHAP feature importance
Left Body 3 - Tail Variability
Right Body 3 - Tail Variability
Angular Velocity
Right Body 1 Movement
Distance between Left Body 2 and Right Body 2
Global Turning Speed
Right Head 1 - Nose Variability
Right Body 1 - Right Body 2 Variability
Center Body Speed
Turning Speed Variability
Left Body 1 Movement
Average Nose Speed
Right Body 1 - Center Distance
Center Speed Variability
Right Head 2 - Tail Distance
Left Body 3 - Center Variability
Body Width Variability
Right Head 1 Lateral Variability
Average Tail Speed
Instant Tail Speed
Freeze
Sniff
Groom
Turn
Locomotion
Climb
Jump
Unassigned
Mean |SHAP value|
C
Cross-validated confusion matrix
True label
Predicted label
D
Held-out cohort
Behaviour dynamics
Freeze dynamics only
ROC AUC
Sanguino Gómez & Krugers → Sanguino Gómez et al.
Sanguino Gómez et al. → Sanguino Gómez & Krugers
E
Full-session cross-validation
ROC AUC
Behaviour dynamics
Freeze dynamics only
F
Full-session contributions
↑ toward resilient
↓ toward vulnerable
Mean |SHAP value|
G
Held-out cohort
Sanguino Gómez & Krugers → Sanguino Gómez et al.
Sanguino Gómez et al. → Sanguino Gómez & Krugers
ROC AUC
Freeze frequency
Sniff frequency
Groom frequency
Turn frequency
Locomotion frequency
Climb frequency
Jump frequency
All frequencies
Simpson index
Shannon entropy index
Evenness index
Cumulative usage index
All diversity indices
Lempel-Ziv complexity
Recurrence rate
Determinism
Markov entropy
All transition metrics
Mean bout duration
Freeze bout duration
Sniff bout duration
Groom bout duration
Turn bout duration
Locomotion bout duration
Climb bout duration
Jump bout duration
All bout durations
All parameters
H
Full-session cross-validation
ROC AUC
I
Behaviour dynamics over time
All behaviours
Freeze
Sniff
Groom
Turn
Locomotion
Climb
Jump
ROC AUC
Time (minutes)
Best predictor overall
Best predictor first 3 min
J
Frequency/diversity/transition/bout duration metrics over time
ROC AUC
Time (minutes)
All frequencies
All diversity indices
All transition metrics
All bout durations
All predictions
K
Individual frequencies
L
Individual diversity indices
Simpson index
Shannon entropy
Evenness index
Cumulative usage
M
Transition metrics
Lempel-Ziv
Recurrence
Determinism
Markov entropy
N
Bout durations
Mean all behaviours